ASYMPTOTICS OF THE SQUARED SZEGOE FUNCTION DIVIDED BY THE WEIGHT AT THE TWO ENDPOINTS

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Allowed standard prerequisites (editorial).

- The asymptotic expansion of the squared Szegoe function divided by the weight at the endpoint one. published statement cited by the author at proof line 475. Supplies the first asymptotic formula of the statement, which the author quotes instead of rederiving it.
- The explicit integral representation of the Szegoe function for a weight on the interval. derived inside the proof at proof lines 477-486. Provides the representation whose exponential argument is analysed to obtain the behaviour at the endpoint minus one.
- The residue computation transferring a Cauchy integral along the cut to a contour integral. derived inside the proof at proof lines 489-497. Rewrites the exponential argument in the form that exhibits the constant defined by the contour integral in the statement.

Source reference. Original reference internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 471–471. The proof refers to the contour depicted in the figure between the statement and the proof by its caption label; that figure is the author's own and is not part of either counted span, and every step of the counted proof is retained verbatim.

Proposition 2.2. *The function* $\frac{F^2}{w}\colon \mathbb{C}\setminus[-1,\infty)\to\mathbb{C}$ *satisfies*

$$\frac{F^2}{w}(z) = 1 \mp \frac{\mathrm{i}\pi}{w(z)} - \frac{\pi^2}{2w^2(z)} + O\left(\frac{1}{\log^3(z-1)}\right) \tag{2.1}$$

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uniformly for any path $z \to +1$ in $\mathbb{C} \setminus [-1, \infty)$, where the $-$ is taken for $\operatorname{Im}(z) > 0$ and $+$ for $\operatorname{Im}(z) < 0$, and also

$$\frac{F^2}{w}(z) = \phi(z)^{-1} \exp\{-(z^2-1)^{1/2}(d_0 + O(|z+1|))\}, \tag{2.2}$$

uniformly for any path $z \to -1$ in $\mathbb{C} \setminus [-1, \infty)$, where

$$d_0 = \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{\log\left(w(\zeta)/(1+\zeta)\right)}{(\zeta^2-1)^{1/2}} \frac{\mathrm{d}\zeta}{\zeta+1} \tag{2.3}$$

and γ is an oriented contour originating from the point $\zeta = 1$, going anticlockwise around the interval $[-1, 1]$ and ending again at the point $\zeta = 1$ as depicted in Figure internal.

Proof. For the proof of statement (2.1), see [1, Proposition A.1]. Statement (2.2) is a special case of [2, Lemma 6.6], but with an additional restriction on the choice of the contour γ stemming from the logarithmic singularity of w. Due to this technicality, we repeat the proof found therein.

First, let us note that if F_1 is the Szegő function of a weight w_1 and F_2 the Szegő function of a weight w_2, then the Szegő function F_{12} of the product $w_1 w_2$ is given by the product of the individual Szegő functions: $F_{12} = F_1 F_2$. As the Szegő function of a Jacobi weight $w_{\alpha,\beta}(x) = (1-x)^\alpha (1+x)^\beta$ with $\alpha, \beta > -1$, is given by (see [2, Remark 5.1])

$$F_{\alpha,\beta}(z) = \frac{(z-1)^{\alpha/2}(z+1)^{\beta/2}}{\phi(z)^{\frac{\alpha+\beta}{2}}},$$

we can conclude that

$$F(z) = \frac{(z+1)^{1/2}}{\phi(z)^{1/2}} \times \exp\left(\frac{(z^2-1)^{1/2}}{2\pi} \int_{-1}^{1} \frac{\log(w(s)/(1+s))}{\sqrt{1-s^2}} \frac{\mathrm{d}s}{z-s}\right), \qquad z \in \mathbb{C} \setminus [-1,1]. \tag{2.4}$$

Again, $(z+1)^{1/2}$ and $\phi(z)^{1/2}$ denote the principal branches (a simple calculation shows that $\phi(z) \notin (-\infty, -1]$ for $z \in \mathbb{C} \setminus (-\infty, 1]$). Moreover, for $x < -1$, $\left(\phi(x)^{1/2}\right)_+ = -\left(\phi(x)^{1/2}\right)_-$ as $\phi(z) \sim 2z$ for $z \to \infty$. Thus $(z+1)^{1/2}/\phi(z)^{1/2}$ is indeed analytic in $\mathbb{C} \setminus [-1, 1]$.

To analyze the argument of the exponential in (2.4), we choose for a fixed $z \in \mathbb{C} \setminus [-1, 1]$ a contour γ as shown in Figure internal. Then a residue calculation shows that

$$\frac{1}{\pi} \int_{-1}^{1} \frac{\log(w(s)/(1+s))}{\sqrt{1-s^2}} \frac{\mathrm{d}s}{z-s} = \frac{\log(w(z)/(1+z))}{(z^2-1)^{1/2}} - \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{\log(w(\zeta)/(1+\zeta))}{(\zeta^2-1)^{1/2}} \frac{\mathrm{d}\zeta}{\zeta - z}. \tag{2.5}$$

Here we note the $w(z)/(1+z)$ is analytic and non-zero in the simply connected region $\mathbb{C} \setminus [1, \infty)$, and hence $\log(w(z)/(1+z))$ exists and is analytic in this region. Moreover, the integrand in (2.5) is integrable along γ as long as $z \notin \gamma$.

Note also that as $\log(w(\zeta)/(1+\zeta))$ has an iterated logarithmic singularity at $\zeta = 1$, we cannot deform γ away from this point. Plugging (2.4) and (2.5) into the definition of $\frac{F^2}{w}$, we obtain

$$\frac{F^2}{w}(z) = \phi(z)^{-1} \exp\left(-\frac{(z^2-1)^{1/2}}{2\pi \mathrm{i}} \int_\gamma \frac{\log(w(\zeta)/(1+\zeta))}{(\zeta^2-1)^{1/2}} \frac{\mathrm{d}\zeta}{\zeta - z}\right).$$

We compute the Taylor expansion around $\zeta = -1$ to obtain

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{\log(w(\zeta)/(1+\zeta))}{\left(\zeta^2-1\right)^{1/2}} \frac{\mathrm{d}\zeta}{\zeta - z} = \sum_{k=0}^{\infty} d_k (z+1)^k,$$

where d_k is given by

$$d_k = \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{\log\left(w(\zeta)/(1+\zeta)\right)}{(\zeta^2-1)^{1/2}} \frac{\mathrm{d}\zeta}{(\zeta+1)^{k+1}}.$$

This finishes the proof. □

References

[1] Conway T.O., Deift P., Asymptotics of polynomials orthogonal with respect to a logarithmic weight, *SIGMA* **14** (2018), 056, 66 pages, arXiv:1711.01590.

[2] Kuijlaars A.B.J., McLaughlin K.T.-R., Van Assche W., Vanlessen M., The Riemann–Hilbert approach to strong asymptotics for orthogonal polynomials on $[-1, 1]$, *Adv. Math.* **188** (2004), 337–398, arXiv:math.CA/0111252.